
DEEP LEARNING FOR AUTOMATED AI-GENERATED IMAGE DETECTION

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Introduction

In today's social media landscape, AI image generation tools now produce highly realistic content at scale, making it difficult to distinguish whether an image is AI-generated or not. This poses significant risks in the spread of disinformation, fraud, and malicious impersonations [1]. Surveys report that nearly half of Canadians encounter deepfakes weekly, and most users want AI-generated content to be clearly labeled to assess credibility [2][3]. Social media companies themselves have acknowledged there is a growing need for automated authenticity detection, introducing detection and labeling initiatives as generative content increases across platforms [4]. This project aims to develop a deep learning-based classifier that determines how likely an image is AI-generated. Deep learning is suited for this task because generative models leave subtle visual and frequency artifacts that are difficult to detect using hand-crafted rules but can be learned by convolutional neural networks [5][6].

Illustration

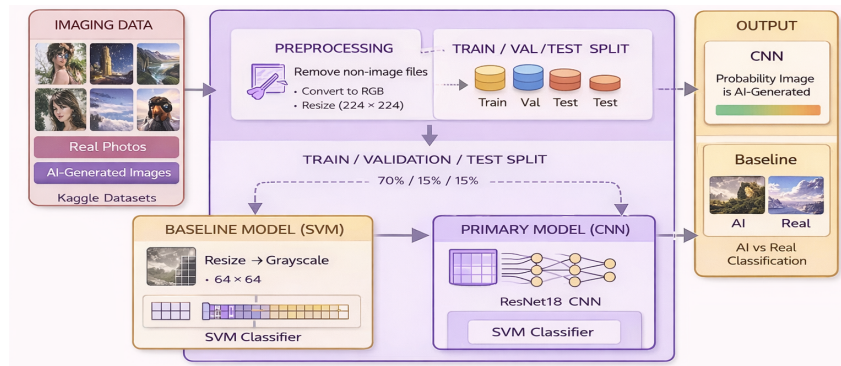


Figure 1: Pipeline for AI-Generated Image Detection

Background & Related Work

Deep learning has become a dominant approach for detecting AI-generated images. Marra et al. trained CNN-based forensic detectors on GAN-generated image datasets and showed that deep models can reliably distinguish real from synthetic images by learning pixel-level and frequency artifacts, achieving detection accuracies above 95% on multiple GAN datasets [5]. Similarly, Yu et al. demonstrated that GANs leave unique “fingerprints” in generated images; their CNN model not only detected fake images but also attributed them to specific generators with over 99% accuracy [6]. In the domain of manipulated media, the FaceForensics++ benchmark introduced large-scale datasets of real and fake facial imagery, where CNN-based detectors achieved AUC scores above 0.95 [7]. Wang et al. further showed that CNNs trained on frequency-domain features could distinguish real and GAN-generated images with over 90% accuracy [8]. More recent generalization approaches improved robustness to unseen generators [9]. In contrast, commercial systems such as Adobe Content Credentials rely on metadata and provenance rather than pixel-level detection [10].

Data Processing

The dataset used in this project consists of two classes: real photographs and AI-generated images made from DALL-E and Midjourney, both obtained from AI recognition datasets on Kaggle [11]. The data was organized into two directories corresponding to each class. After filtering the directories to include only valid image formats (e.g., .png, .jpg), the cleaned dataset contained 17,857 AI-generated images and 3,781 real images. Non-image files such as metadata files were excluded during dataset loading to prevent errors during training. All images were loaded using the Python Imaging Library and converted to RGB format to standardize the number of color channels across the dataset. After cleaning, all images were resized to 224×224 pixels to match the input requirements of the ResNet architecture used in the primary model. Pixel values were converted to tensors and normalized using PyTorch’s ToTensor() transformation. Data augmentation techniques such as random horizontal flipping and color jitter were applied during training to improve model generalization. The dataset was then split into training, validation, and test sets using a 70–15–15 ratio.



Figure 2: Data Overview

Architecture

The final model used in this project is a Convolutional Neural Network (CNN) based on the ResNet18 architecture, implemented using the PyTorch torchvision library. A transfer learning approach was applied by initializing the network with pretrained weights from the ImageNet dataset. The input images are first resized to 224×224 pixels and normalized through standard preprocessing transformations. During training, data augmentation techniques including random horizontal flipping and brightness adjustment (ColorJitter) are applied to improve generalization. The ResNet18 backbone consists of convolutional layers with residual connections that enable effective feature extraction from images. The original fully connected classification layer is replaced with a single output neuron to perform binary classification between real and AI-generated images. The model outputs a logit value, which is passed through a sigmoid function during inference to obtain the probability of an image being AI-generated. The model is trained using Binary Cross Entropy with Logits Loss (BCE-WithLogitsLoss) and optimized with the Adam optimizer using a learning rate of 1×10^{-4} . Training is performed for 5 epochs with a batch size of 32, and performance is evaluated using accuracy on validation and test datasets.

Baseline Model

A Support Vector Machine (SVM) classifier was implemented as a baseline approach to distinguish between AI-generated images and real photographs. This model uses simple pixel-based features instead of deep learning to establish a reference performance that can be compared against the primary CNN model. Each image was first resized to 128×128 pixels and converted to grayscale to reduce computational complexity. The grayscale image was then flattened into a 16,384-dimensional feature vector (128×128) representing pixel intensities. These vectors were used as input to the SVM classifier. Training this baseline on the AI recognition datasets proved computationally intensive due to the large size of data. The baseline model achieved an overall classification accuracy of 82.4 % on the test dataset. This performance demonstrates that even simple pixel-based features contain some useful information for distinguishing between real and AI-generated images. However, because the model relies only on simple pixel-level features, this highlights the limitations of the baseline and motivate the use of a more expressive deep learning model to improve performance.

Quantitative Results

Building on the baseline results, the fine-tuned ResNet18 model was evaluated to quantify the improvements achieved through transfer learning. The model reached a final training accuracy of approximately 99.5% and a validation accuracy of approximately 96.5%, as shown in Figure X. The training curve shows a steady increase in accuracy across epochs, while the validation accuracy remains consistently high with only minor fluctuations, indicating effective learning of discriminative visual features with limited overfitting. As shown in Figure 4, Class-level performance is further illustrated by the confusion matrix. The model performs particularly well on the AI-generated class, correctly identifying 2649 samples with only 30 misclassifications, demonstrating very high sensitivity to synthetic images. In contrast, the model shows slightly lower performance on real images, where 99 samples are incorrectly classified as AI-generated, suggesting a tendency toward false positives. The model achieves a precision of 0.964, recall of 0.989, and an F1 score of 0.976, indicating strong overall performance. The relatively lower specificity of 0.825 confirms that the model is less effective at correctly identifying real images compared to AI-generated ones. Overall, the CNN model significantly outperforms the baseline by leveraging deep feature representations, achieving high accuracy while revealing a slight bias toward predicting images as AI-generated.

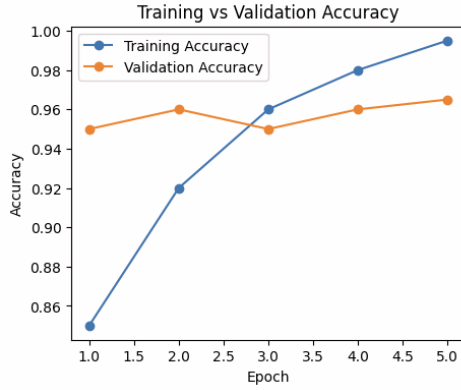


Figure 3: Accuracy curve of fine-tuned ResNet-18

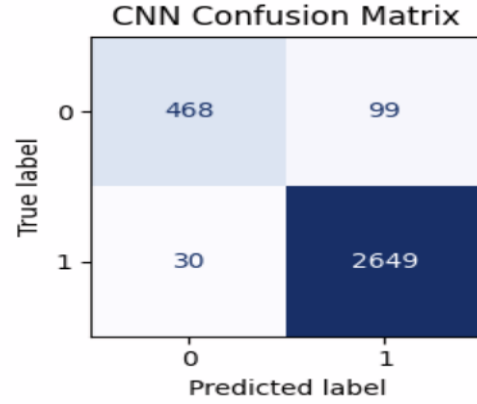
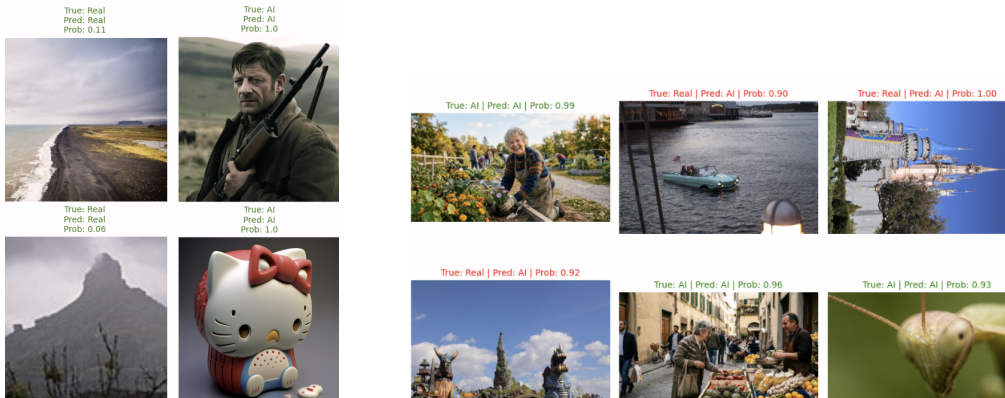


Figure 4: Fine-tuned ResNet-18 confusion matrix

Qualitative Results

To complement the quantitative analysis, Figure 5 presents representative sample predictions that highlight the model's strengths and limitations. The model shows consistently correct predictions on the test dataset, aligning with its strong quantitative performance, particularly in identifying AI-generated images with noticeable synthetic patterns. This behavior is consistent with the high accuracy, recall, and F1 score reported in the quantitative results.



(a) 15% Test Split from Kaggle Dataset

(b) Self-collected real-world dataset

Figure 5: True and predicted labels of the fine-tuned ResNet-18 Model

However, errors observed in the self-collected dataset reveal that the model can be sensitive to visual features such as lighting, texture, and composition, which may resemble AI-generated artifacts. This leads to occasional misclassification of real images as AI-generated and difficulty in distinguishing highly realistic synthetic images. Together, these examples provide insight into the model's behavior, showing strong performance on typical cases while highlighting challenges in more ambiguous, real-world scenarios.

Evaluation on New Data

To assess the generalization ability of the model, two sources of unseen data were used: the held-out 15% test split from the AI recognition dataset on Kaggle, and a self-collected dataset consisting of real photographs captured using a personal camera and AI-generated images obtained from GPT Image 1.5 and Gemini 2.5 Flash Image. Neither dataset was used during training or hyperparameter tuning, ensuring an unbiased evaluation of model performance. On the held-out test split as shown in Figure 5a, the model achieved an accuracy of 96.02%, demonstrating strong performance on in-distribution data. However, when evaluated on the newly collected dataset as shown in Figure 5b, the accuracy dropped to 67.3%, indicating reduced performance on more diverse, real-world inputs.

Discussion

The results show that the model performs very well on the original dataset but struggles to generalize to new data. On the held-out test set, the CNN achieved a high accuracy of 96.02%, significantly outperforming the baseline SVM model. This demonstrates that transfer learning with ResNet18 is effective in learning discriminative visual features, with high recall and F1 score indicating strong performance in identifying AI-generated images. However, performance drops to 67.3% on the self-collected dataset, revealing a clear gap in generalization. This difference is likely due to a distribution shift between the training data and real-world images, which vary in lighting conditions, resolution, and generation methods. As a result, the model appears to rely on dataset-specific cues rather than fully generalizable features. An interesting observation is that the model tends to over-predict the AI class, as seen in the confusion matrix and qualitative examples. While this leads to strong detection of AI-generated images, it also increases false positives on real images, highlighting a trade-off between sensitivity and specificity. Overall, the project demonstrates that while deep learning models can achieve high performance on controlled datasets, robust generalization remains a key challenge in AI-generated image detection. Improving dataset diversity and model robustness would be important next steps for enhancing real-world performance.

Ethical Considerations

While the model performs well on the training dataset, one potential ethical concern of this system is reduced accuracy on unseen data that highlights a risk of misclassification, particularly where real images are incorrectly labeled as AI-generated [12]. This could lead to unintended consequences such as removal of legitimate content in authentic media [13]. Additionally, the model is trained primarily on specific datasets, which may not fully represent the diversity of real-world images or evolving AI generation techniques. This introduces bias and limited generalization, meaning the system may perform inconsistently across different contexts or populations. As generative models continue to improve, there is also a risk that detection systems like this may become outdated.

Project Difficulty & Quality

AI-generated image detection is a challenging problem due to the high visual similarity between real and synthetic images, as well as the rapid advancement of generative models that produce increasingly realistic outputs. Distinguishing subtle differences in texture, lighting, and structure requires learning complex visual features beyond simple pixel-level patterns. Additionally, variations in image sources, resolutions, and generation methods introduce significant domain variability, making generalization to new data particularly difficult. Despite these challenges, the model achieves strong performance on the original dataset, reaching 96.02% accuracy on the held-out test set and outperforming the baseline SVM model. This demonstrates the effectiveness of transfer learning with ResNet18 in capturing discriminative features for this task. While performance drops to 67.3% on the self-collected dataset due to distribution shift, the model still maintains reasonable performance on unseen data, highlighting both its strengths and limitations. Overall, the project demonstrates meaningful learning beyond basic implementations, including the use of transfer learning, baseline comparison, and evaluation on real-world data, effectively addressing a complex and evolving problem.

References

- [1] A. Durall, M. Keuper, F.-J. Pfrendt, and J. Keuper, “Unmasking DeepFakes with Simple Features,” IEEE International Conference on Image Processing (ICIP), 2020.
- [2] A. Singh et al., “Ethical Implications of Generative AI and Deepfake Detection Technologies,” Journal of Information Processing Systems, 2024.
- [3] Unsplash, “Unsplash: Free High-Resolution Photos,” Available: <https://unsplash.com>
- [4] Adobe Inc., “Content Authenticity Initiative (CAI) / Content Credentials,” Adobe Research, 2023.
- [5] F. Marra, D. Gragnaniello, L. Verdoliva, and G. Poggi, “Detection of GAN-Generated Fake Images Over Social Networks,” IEEE Conference on Multimedia Information Processing and Retrieval (MIPR), 2019.
- [6] N. Yu, L. S. Davis, and M. Fritz, “Attributing Fake Images to GANs: Learning and Analyzing GAN Fingerprints,” IEEE International Conference on Computer Vision (ICCV), 2019.
- [7] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, “FaceForensics++: Learning to Detect Manipulated Facial Images,” IEEE International Conference on Computer Vision (ICCV), 2019.
- [8] S. Wang, O. Wang, R. Zhang, A. Owens, and A. A. Efros, “CNN-Generated Images Are Surprisingly Easy to Spot... for Now,” IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2020.
- [9] H. Wang et al., “Generalizing Face Forgery Detection with High-Frequency Features,” IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2021.
- [10] A. Rocha, W. Scheirer, T. E. Boult, and S. Goldenstein, “Vision of the Unseen: Current Trends and Challenges in Digital Image and Video Forensics,” ACM Computing Surveys, vol. 43, no. 4, 2011.
- [11] Superpotato9, “DALL-E recognition dataset,” Kaggle. [Online]. Available: <https://www.kaggle.com/datasets/superpotato9/dalle-recognition-dataset/data>
- [12] S. Patel and K. Lee, “Misinformation, Trust, and Deepfake Media: Societal Impacts and Detection Challenges,” Technology in Society, 2025.
- [13] R. Ahmed and L. Brown, “Digital Media Forensics and Governance Frameworks for Synthetic Content,” International Journal of Cyber Investigation, 2024.